

AI

(Intro to Artificial Intelligence)



MONTRÉAL, QUEBEC, CANADA



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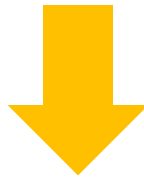


BEAST

AI Content Generation

Generating images/video from
natural language:

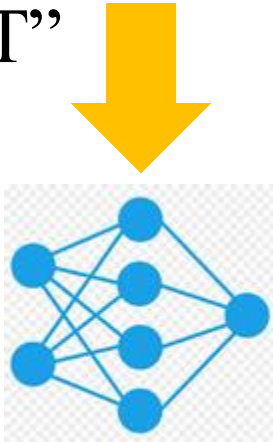
“A snowy view of Old Port
in Montreal”



AI Content Generation

Generating programs from
natural language:

“write code in tensorflow
to train a model using
MNIST”



```
Sure! Here's a simple TensorFlow code to train a model using the MNIST dataset. This code uses the tf.keras API, which is the high-level interface in TensorFlow for building and training models.

python Copy code

import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical

# Load MNIST dataset
(x_train, y_train), (x_test, y_test) = mnist.load_data()

# Preprocess the data
x_train = x_train.reshape(x_train.shape[0], 28, 28, 1) # Reshape to (28, 28, 1)
x_test = x_test.reshape(x_test.shape[0], 28, 28, 1) # Same for test set
x_train, x_test = x_train / 255.0, x_test / 255.0 # Normalize pixel values

# One-hot encode the labels
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Build the model
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(1024, activation='relu'),
    layers.Dense(1024, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test))
```

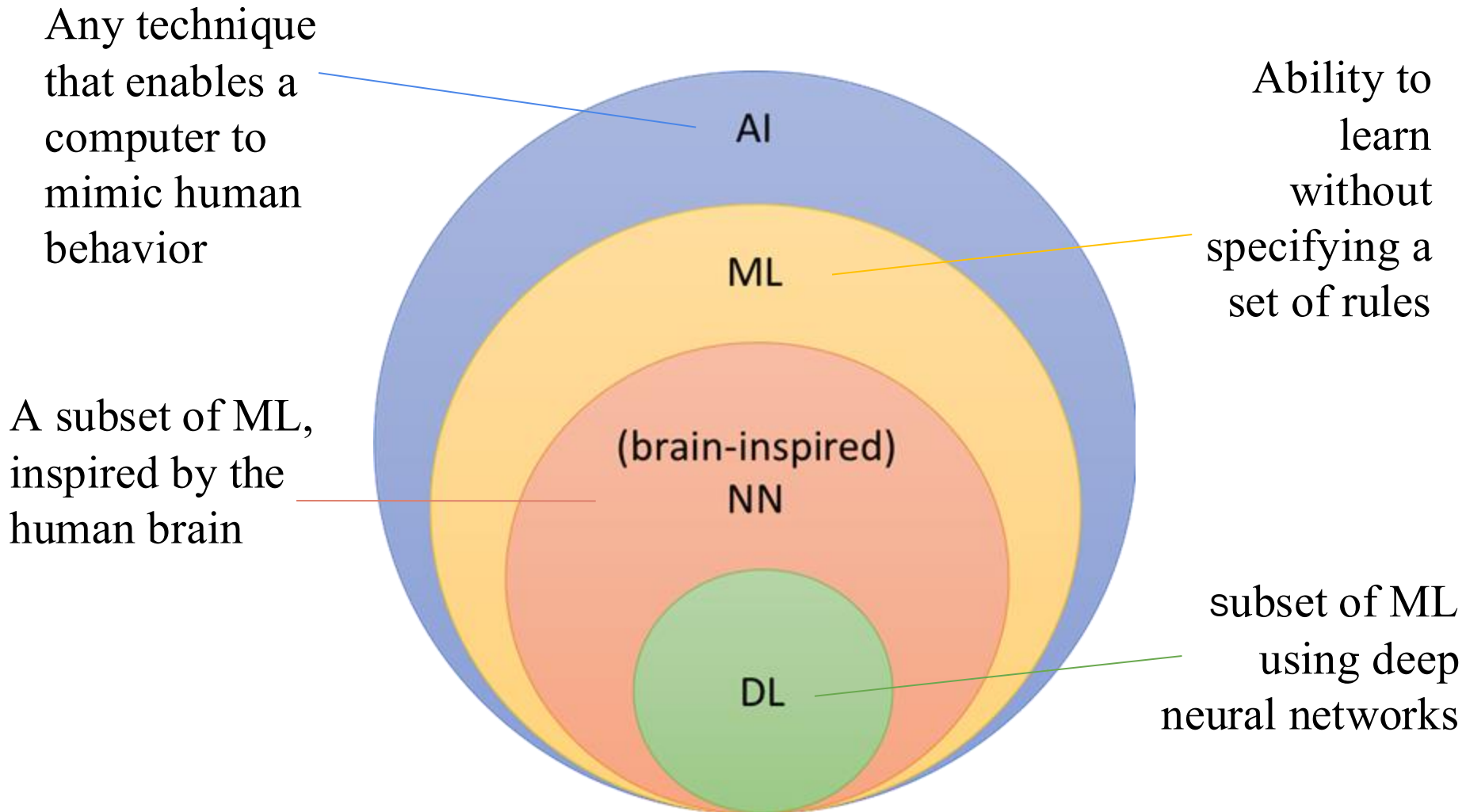
Message ChatGPT

ChatGPT can make mistakes. Check important info.

Where is AI heading?

- Artificial Intelligence (AI) is seamlessly integrating with our everyday life activities.
- Sometimes, we are not even aware of its use.

What is AI?

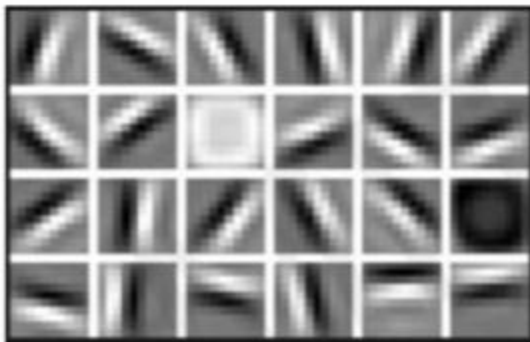


Why Deep Learning?

ML requires hand engineered features, which are expensive and time consuming to design.

Could we learn these features directly from the data?

Low Level Features



Lines & Edges

Mid Level Features



Eyes & Nose & Ears

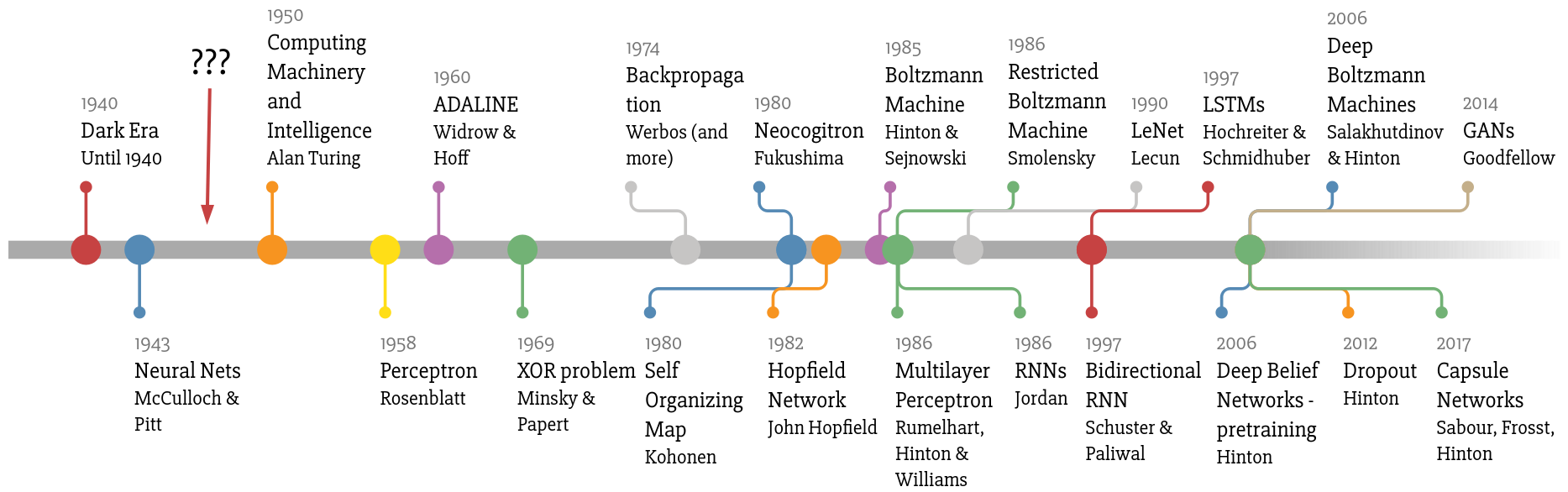
High Level Features



Facial Structure

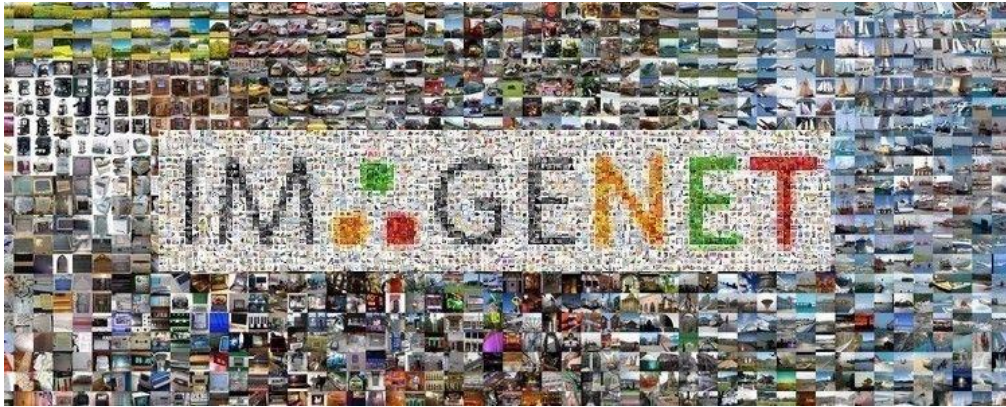
Why Now?

- Neural networks are not new. They date back decades ago.

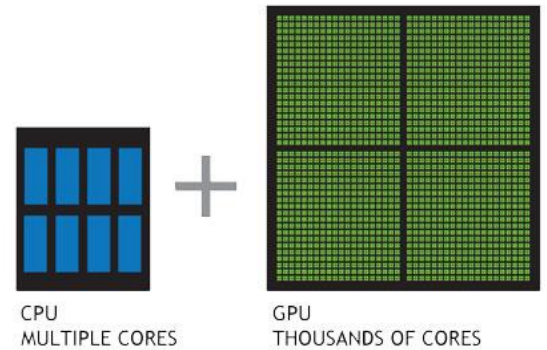


Why Now?

- so why the sudden dominance these days?



Abundance of data



Advanced Hardware

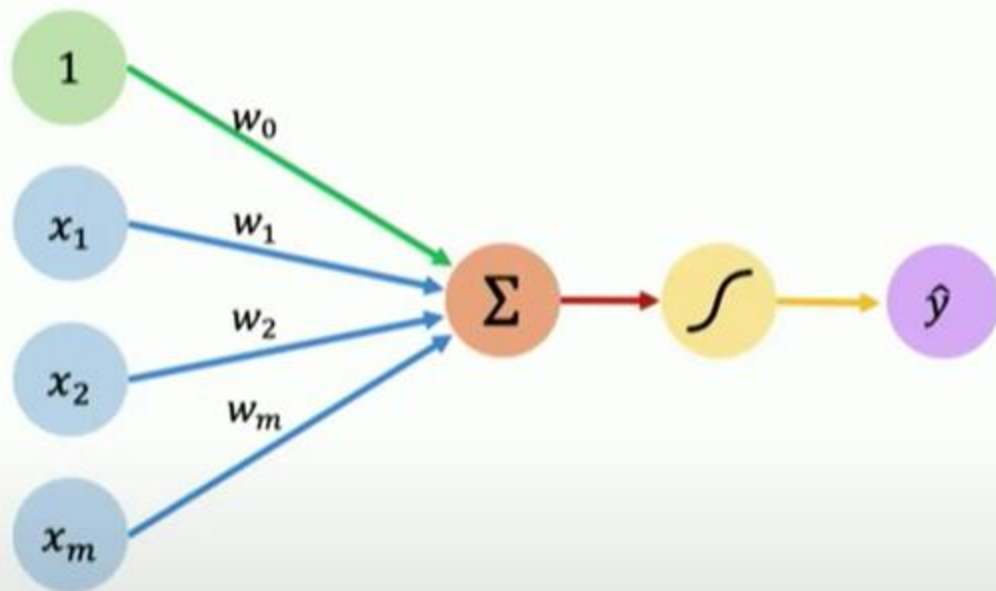


Advanced Programming Languages



How AI learns?

The Perceptron: Forward Propagation



Inputs Weights Sum Non-Linearity Output

Output

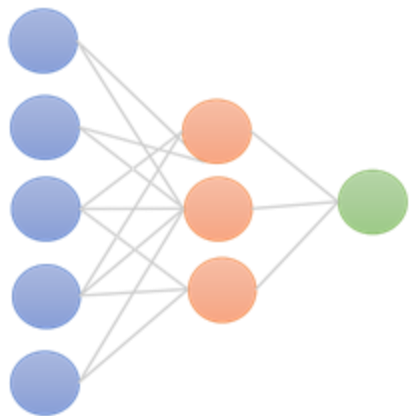
Linear combination of inputs

$$\hat{y} = g \left(w_0 + \sum_{i=1}^m x_i w_i \right)$$

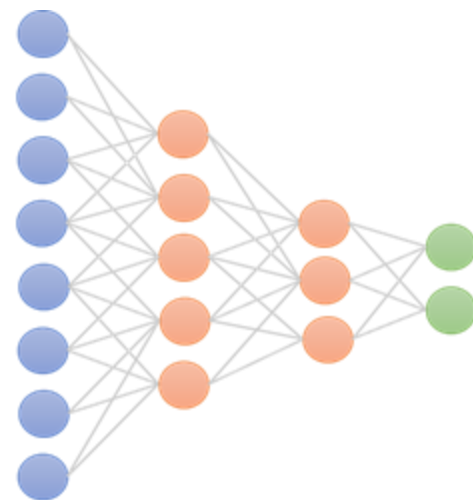
Non-linear activation function

Bias

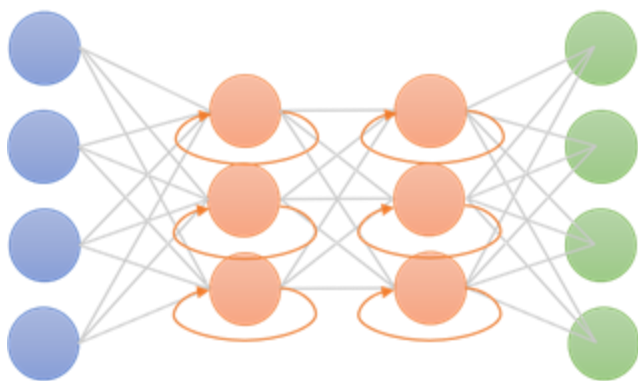
Types of NN



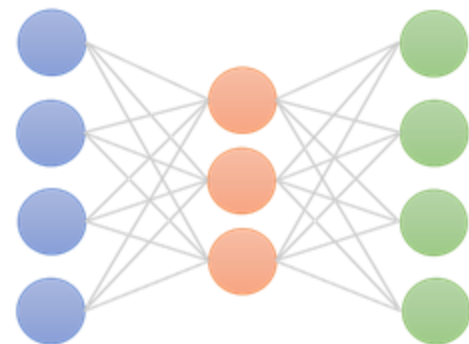
a) Feed-forward neural network



b) Convolutional Neural Network



c) Recurrent Neural Network



d) Autoencoders

Input Data

Data preprocessing is the process of:

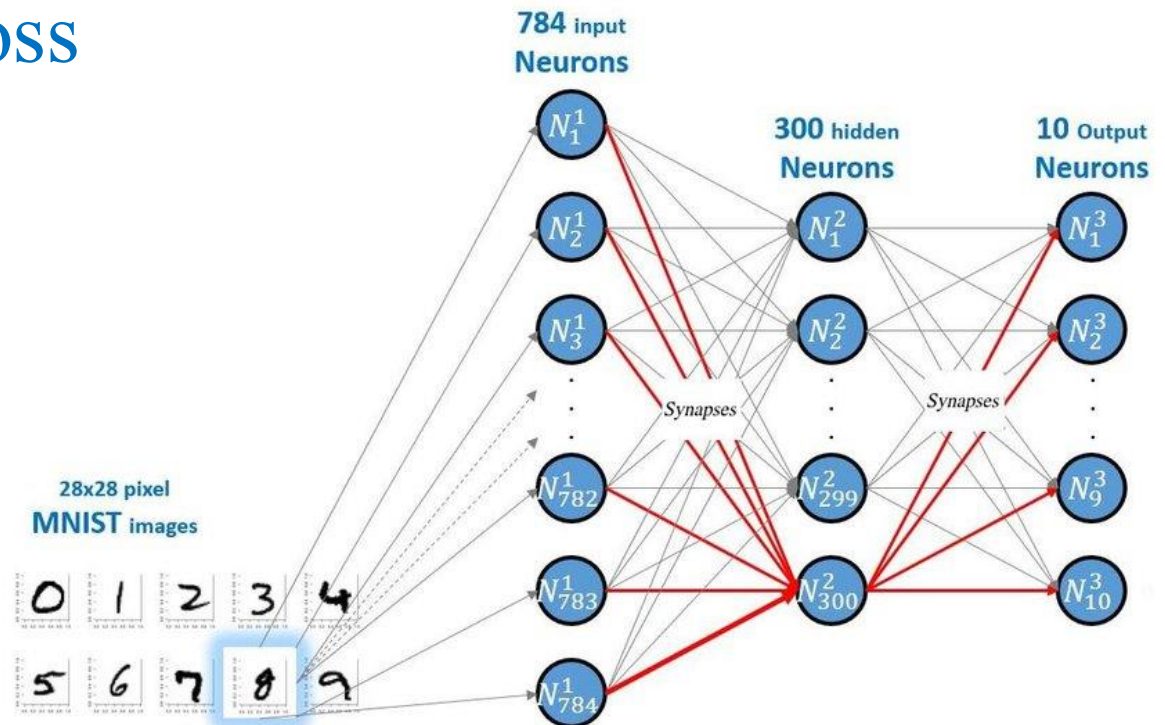
- evaluating,
- filtering,
- manipulating, and
- encoding data

The major goal of data preprocessing is to eliminate data issues:

- missing values,
- improve data quality,
- and make the data useful for machine learning purposes.

Training Neural Networks

The purpose of **training** neural nets is to find the optimal network weights that achieve the **lowest loss**



Loss Functions ?

Genera





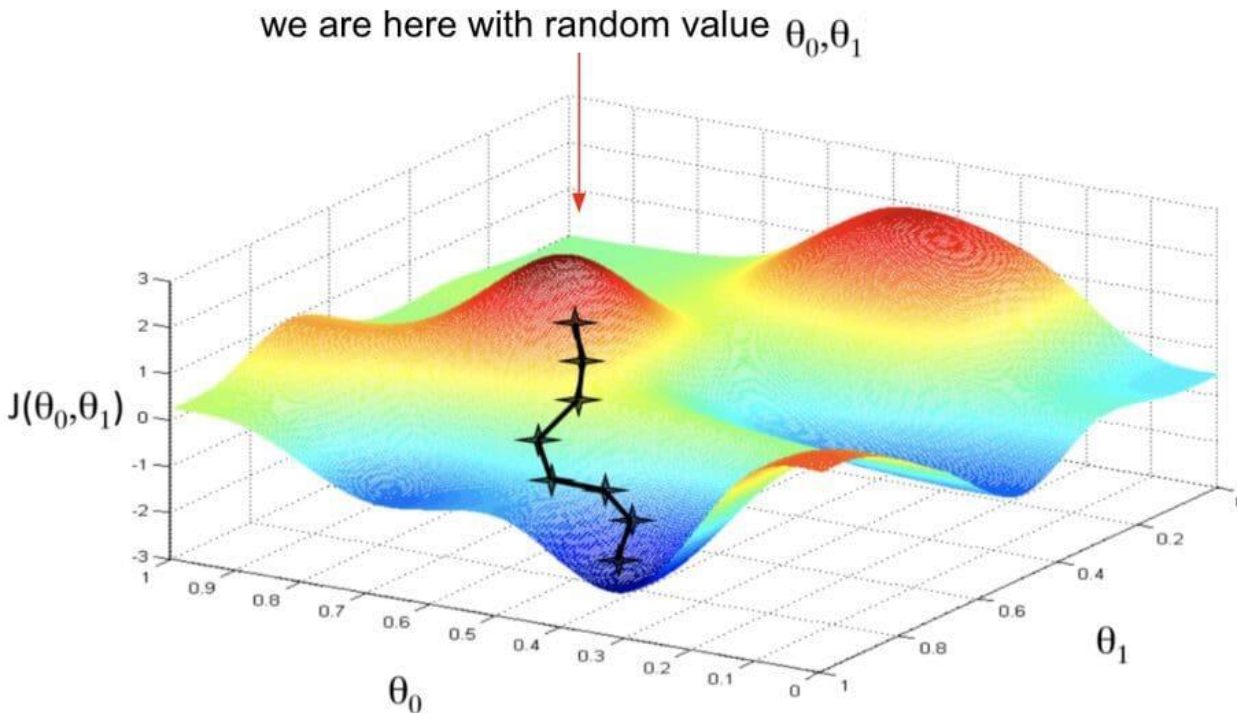
The lowest loss?



Player 1

Player 2

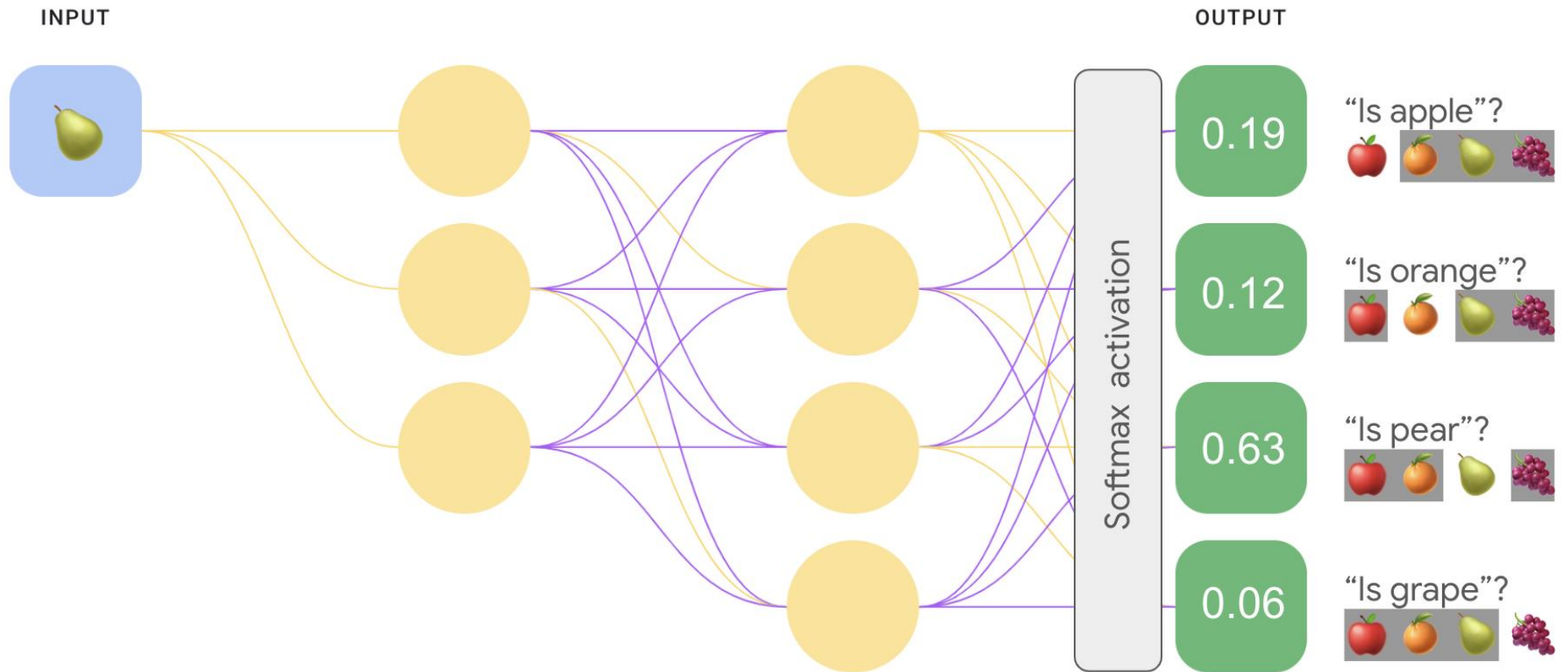
Loss function



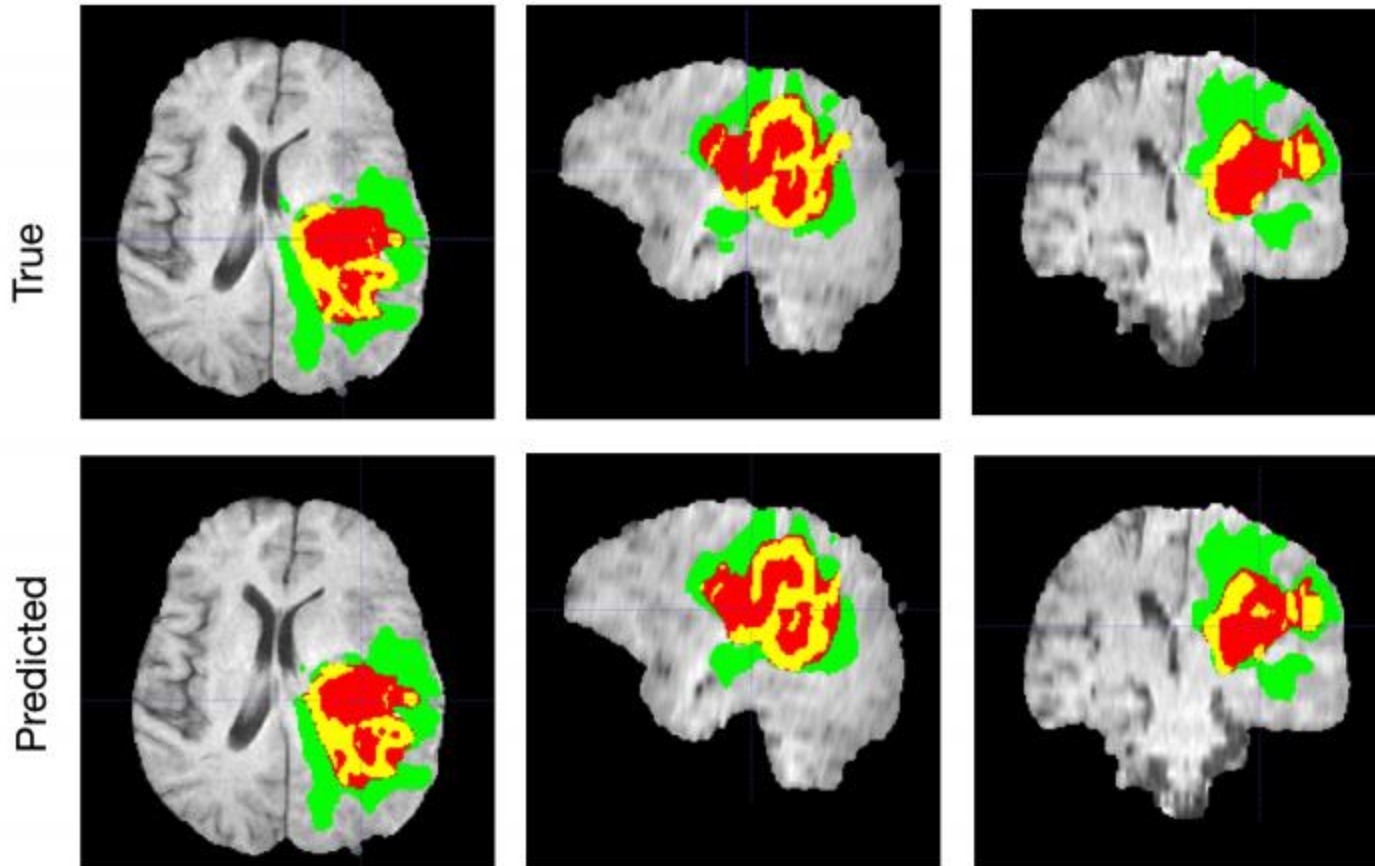
- Start with some θ_0, θ_1
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we hopefully end up at a minimum

Gradient Descent
(CALC I. Derivatives)

Output



Brain Tumor Diagnosis



Other cool applications

Source: [Kim et al. 2015]

